

Rodrigo Morais

SENIOR SOFTWARE AND DATA PLATFORM ENGINEER

Göttingen, Germany | Remote preferred | Hybrid nearby | No relocation
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Professional Summary

Senior Software and Data Platform Engineer with 30 years of professional experience, including more than seven years specializing in data engineering. Builds distributed systems, streaming and CDC architectures, production data platforms, and reliable cloud services. Recent results include 10x faster processing, 34% lower Snowflake cost, and 72% fewer production data incidents.

Technical Skills

Software	Python, Go, TypeScript, JavaScript, Java, Clojure, C#, Ruby, APIs, distributed systems, event-driven design, CQRS
Data Platforms	PySpark, Spark Structured Streaming, Kafka, Snowflake, dbt, PostgreSQL, Cassandra, CDC, data contracts
Orchestration	Prefect, Airflow, recoverable workflows, idempotency, retries, backfills, monitoring, reconciliation
Cloud and Platform	AWS, Docker, Kubernetes, Terraform, CI/CD, Prometheus, Grafana
Production AI	LangGraph, typed decisions, model routing, guardrails, automated evaluations, observability

Professional Experience

Independent Software Data and AI Engineer

Germany | August 2025 to present

INVESTMENT MANAGER PERSONAL INVESTMENT MANAGEMENT PLATFORM

- Built a private software platform for quantitative research, portfolio operations, market data, strategy evaluation, and controlled decision support.
- Increased data-processing performance by 10x by replacing batch pipelines with PySpark streaming pipelines.
 - Increased usable market-data coverage from 94% to 99.8% through multi-source ingestion, automatic fallback, validation, reconciliation, and recovery.
 - Reduced portfolio reconciliation and reporting from 90 minutes to under 10 minutes by automating document ingestion, position reconciliation, calculations, and validation.
 - Reduced quantitative-research cycles from two days to under two hours by converting exploratory notebooks and time-series studies into reproducible production pipelines.
 - Reduced AI operating costs by 60% through domain-specific agents, graph workflows, model routing, guardrails, reference datasets, evaluations, and production observability.

Data Engineer

Remote | February 2021 to July 2025

JUSTWATCH GMBH

- Engineered production data systems for high-volume product, analytical, and operational workloads.
- Accelerated end-to-end data processing by 4x through automated ingestion, validation, testing, orchestration, and recovery.
 - Reduced Snowflake costs by 34% by optimizing pipelines and SQL, removing redundant workloads, and moving suitable preparation upstream to PostgreSQL and Cassandra.
 - Moved business-critical refreshes from twice daily to near real time using PostgreSQL change data capture and Spark Structured Streaming into Snowflake.
 - Reduced production data incidents by 72% through observability, infrastructure improvements, configuration hardening, and automatic recovery.
 - Increased analyst productivity by 50% through automated tests, CI/CD, and shared data infrastructure.

Professional Experience Continued

Quantitative Finance Developer

Remote | May 2020 to February 2021

TC TRADERS CLUB

Modernized market-data collection and analytical delivery for quantitative research.

- Reduced market-data latency from one day to 15 minutes and increased availability from 95% to 99.5% through validation, retries, monitoring, and recovery.
- Reduced data acquisition and preparation from hours to minutes by introducing Snowflake and dbt for analytical modeling and tests.
- Raised validated market-data correctness to nearly 100% by redesigning collection, cleaning, transformation, and testing workflows.

Senior Software Engineer Data Migration

Remote | January 2019 to May 2020

TRADESHIFT

Built distributed migration capabilities used by teams moving from a monolith to services.

- Developed a CQRS-based migration platform that let more than 10 teams adopt microservices while continuing to use existing PostgreSQL data safely.
- Reduced migration-related production incidents by 70% through idempotent processing, automated validation, replayable migrations, and standardized contracts using Clojure, RabbitMQ, and Docker.

Senior Full Stack Software Engineer

Berlin Germany | November 2016 to January 2019

BABEL

Developed cloud-backed product capabilities and modernized web application architecture.

- Reduced feature deployment time by 20% by automating AWS infrastructure and delivery with Terraform and CI/CD across API Gateway, Lambda, DynamoDB, S3, and RDS.
- Improved application performance by 40% by migrating legacy JavaScript to modular React and Redux applications and separating scalable service responsibilities.

Earlier Engineering and Technical Leadership

Software Engineer Senior Engineer Technical Lead | ThoughtWorks, ITS4, Scopus IT, CWI, DBC and earlier companies | 1997 to 2016

- Led teams of up to six engineers and delivered projects for international clients including Gap, LATAM, Bradesco, and industrial organizations.
- Built data warehouses, reporting platforms, migration services, REST APIs, distributed applications, CI/CD pipelines, and modernization solutions across banking, ERP, industrial production, and web systems.

Education

M.Sc. Software Engineering | UNIBTA Instituto Brasileiro de Tecnologia Avançada | 2010

Service Oriented Architecture

B.Sc. Applied Mathematics | ULBRA Universidade Luterana do Brasil | 2008

Applied Mathematics for Computer Science

Languages

Portuguese native | English professional | Spanish intermediate | German A1